Department of Artificial Intelligence & Data Science

School of Computing, FAST-NUCES

Degree: DS/AI	**Total Marks:** 20
Instructor: Dr. Zohair Ahmed	**Due Date:** As Per GCR

Assignment 2: Reproduction of Results

Objective

- Develop **technical skills and experimental rigor** by reproducing the results of the selected paper.
- Students should use the **same datasets and evaluation metrics when possible**.
- If any of the provided **papers does not include a proposed model or dataset**, you must inform the teacher before finalizing your selection. If you select such a paper without informing the teacher, **any resulting issues will be the responsibility of the group**.

Important Policy

If a student submits work that appears to be **copied from AI tools or external sources and cannot explain it**, the instructor may assign **zero marks for the project**.

Therefore, please ensure that:

- you actively participate in the project
- you understand the work being submitted
- you maintain academic integrity

Tasks

1. Implementation

Students should:

- Use the **official code repository** if available
- Otherwise implement the model from scratch

Frameworks may include:

- Hugging Face Transformers
- PyTorch
- TensorFlow

---

2. Experimental Setup

Describe clearly:

- Hardware used (GPU/CPU)
- Dataset preprocessing
- Hyperparameters
- Training configuration

---

3. Reproduced Results

Present:

- Performance metrics
- Tables comparing reproduced results vs original paper
- Training curves

Example:

Model	Paper Result	Reproduced Result
Proposed Model	92.4	91.7

---

4. Reproducibility Analysis

Discuss:

- Differences from original results
- Possible reasons for discrepancies
- Implementation challenges
- Dataset or environment differences

Deliverables

Report Length: 8–10 pages

Structure:

1. Introduction
2. Paper Overview
3. Implementation Details
4. Experimental Setup
5. Reproduced Results

6. Reproducibility Discussion
7. Conclusion
8. References

Students must also submit:

- **Code repository**
- **Experiment logs**

Evaluation Criteria (Total 100)

Criteria	Weight
Correct implementation	30%
Experimental rigor	25%
Result comparison	20%
Discussion of challenges	15%
Code quality & documentation	10%
